

# Corentin Cadiou

Assistant professor  
*Chargé de recherche*

J 16/09/1992

H Male

☑ French

☑ Institut d'Astrophysique de Paris (IAP)  
98 bis boulevard Arago  
75014 Paris, France

☑ cphyc.github.io

☑ github.com/cphyc

☑ 0000-0003- 2285-0332

☑ +33 6 43 18 66 83

☑ corentin.cadiou@iap.fr

## Science interests

galaxy formation  
cosmic web  
numerical simulations  
cosmology

## Languages

French (native)

English (C2)

German (B2)

Spanish & Swedish (A1)

## Numerics

### HPC

MPI OpenMP

Kokkos

### Programming

Fortran C++

Python

## Research experience

- 2025–now **Chargé de recherche (Assistant Professor)** IAP, France 
- Permanent, 100%-research position. Recruited on a interdisciplinary project to develop high-performance computing and data science in astronomy.
- 2022–25 **Post-doctoral research** Lund, Sweden 
- Working on the group of Prof. Agertz on the role of angular momentum in the formation of galactic disks. Start: 01/10/2022, end: 31/01/2025
- 2019–22 **Post-doctoral research** UCL, London, UK 
- With Profs. Pontzen and Peiris, on ERC grant.
- 2016–19 **Post-graduate research** IAP, Paris, France 
- Supervisors: C. Pichon and Y. Dubois.

## Education

- 2019 **PhD in Astrophysics** Sorbonne & IAP, Paris  
- “The impact of the large-scale structures of the Universe on dark matter halo and galaxy formation”. Refereed by S. White and A. Dekel.
- 2016 **Master's degree (Master 2) in Astronomy and Astrophysics** Univ. Paris Diderot, Paris Observatory, Paris, France  
- 2015 Diploma of the École Normale Supérieure (ENS) ENS, Paris 
- Major in physics, minor in Computer Sciences
- 2013 Bachelor's degree, Physics Univ. Paris Diderot & ENS, Paris  

## Time allocations

Over my career, I have been **PI or co-I of projects securing 400 MCPU hr** (4,000,000€, assuming a price of 0.01€/CPU hr). My developments also enabled additional projects for a total of more than 100 MCPU hr.

- 2024–now **(co-I) Harkonnens simulations**   
- 250 MCPU hr** (EuroHPC) + **60 MCPU hr** (Spanish national call). Suite of high-resolution simulations to support ESA's ARRAKIHS mission to investigate the nature of dark matter.
- 2024 **(PI) The role of mergers in shaping Milky-Way galaxies** 
- 6 MCPU hr allocation** (Swedish national call). Suite of high-resolution simulations focused on the role played by mergers in the formation of our galaxy.
- 2024 **(PI) How the cosmological environment drives galaxy properties** 
- 3.6 MCPU hr allocation** (local call). Suite of simulations to unravel the role played by the cosmological environment in setting the properties of galaxies.
- 2023–25 **(co-I) MEGATRON project** 
- Large 50 MCPU hr allocation** (UK national call), 15th DiRAC call (PI: H. Katz). Extreme-resolution cosmological simulation focused on circum-galactic physics.
- 2021–22 **(PI) Angular momentum project** 
- 9.7 MCPU hr allocation** (UK national call), 13th DiRAC call. Demonstration of the feasibility of controlling the angular momentum of galaxies in a cosmological volume.
- 2021–24 **EDGE Project** (‘code builder’ status) 
- Automatically co-author of all publications that use my contributed code. 40 MCPU hr obtained (UK national call, PI: J. Read). Suite state-of-the-art simulations of dwarf galaxies.
- 2020–21 **Obelisk simulation** 
- Radiation-hydrodynamical cosmological simulation following the assembly of a proto-cluster. 50 MCPU hr obtained (Europe wide call, PI: M. Trebitsch).
- 2018–20 **CINES computational time allocation** 
- Co-I of a 2 MCPU hr subproject, 25 MCPU hr obtained (France national call, PI: M. Volonteri). Investigation on the role of cosmological accretion on angular momentum accretion.

## Awards and recognitions

|         |                                                                                                                                                                 |                         |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 2024-26 | <b>eSSENCE grant (1 100 000 kr <math>\approx</math> 95 000 €)</b><br>Research grant for the project: “Galaxy formation in the exascale era”.                    | Lund University, Sweden |
| 2024-26 | <b>Fysiografen grant (110 000 kr <math>\approx</math> 9 500 €)</b><br>Research grant for the project: “The formation of disk, from cosmic dawn to cosmic noon”. | Lund University, Sweden |
| 2023-25 | <b>Fysiografen grant (140 000 kr <math>\approx</math> 2 000 €)</b><br>Research grant for the project: “The role of environment in driving galaxy spin”.         | Lund University, Sweden |
| 2018    | <b>NumFOCUS New Contributor Award</b><br>In recognition of my contributions to the Yt project, the most widely-used Python package for analysing simulations.   |                         |
| 2016–19 | <b>ILP fellowship (5000 € per annum)</b>                                                                                                                        |                         |
| 2012–19 | <b>ENS scholarship &amp; ENS doctoral fellowship, prestigious full stipends awarded nationwide to 20 fellows.</b>                                               |                         |

## Responsibilities

### — International collaborations & code development for open-science

|          |                                                                                                                                                                                                                                                                                                             |        |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 2023–now | <b>ARRAKIHS mission</b><br>European Space Agency (ESA) space mission to shed light on the nature of dark matter, to be launched in 2030. Co-I of the Simulation Work Package to interpret the data.                                                                                                         |        |
| 2023–now | <b>‘Agora’ collaboration</b><br>Code comparison project aimed at finding which galaxy properties are robust predictions from the different models.                                                                                                                                                          |        |
| 2022–now | <b>‘Ginea’ collaboration</b><br>Collaboration to develop the next-generation cosmological simulation code (DYABLO, to supersede RAMSES). Personal contributions include key insight into input/output formats and coupling with post-processing tools.                                                      | France |
| 2019–24  | Member of ERC GMGalaxies (2019–2022, PI: Pontzen).                                                                                                                                                                                                                                                          |        |
| 2016–24  | Member of ANR Spine (2016–2017, PI: Pichon) and SEGAL (2019–2024, PI: Pichon).                                                                                                                                                                                                                              |        |
| 2017–now | <b>Yt team member</b> , in charge of support of the RAMSES code.<br>Yt is now the most widely used library to analyse astrophysical simulations. Personal contributions include support for the RAMSES code, significant I/O performance improvements ( $\times 100$ faster for RAMSES), community support. |        |

### — Community service

|          |                                                                                                                                                                                                                                                                                       |                                      |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 2025–now | <b>IAP Seminars</b><br>In charge of the organization of IAP’s seminars (weekly).                                                                                                                                                                                                      |                                      |
| 2022–now | <b>Member of the EAS Advisory Committee on Sustainability</b><br>The European Astronomical Society (EAS) Sustainability Advisory Committee aims to investigate, communicate, and make recommendations to the Council on sustainability matters related to astronomy and astrophysics. |                                      |
| 2020–now | <b>Reviewer for Astronomy and Astrophysics, Monthly Notices of the Royal Astronomical Society, Scipy’s conference proceedings</b>                                                                                                                                                     |                                      |
| 2016–21  | Organizer of IAP pre-seminar and the ‘Extragalactic Journal Club’                                                                                                                                                                                                                     | IAP, Paris, France & UCL, London, UK |

### — Teaching and supervision

|          |                                                                                                      |  |
|----------|------------------------------------------------------------------------------------------------------|--|
| 2025–now | <b>PhD student supervisions</b><br>Supervision of 1 PhD student: S. Errachi (IAP, Master 2, 25–now); |  |
| 2020–now | <b>Master’s student supervisions</b>                                                                 |  |

Supervision of 9 Master's students. The work of the students in bold led to a submitted paper:

- Y. Su (Master Calcul Haute Performance et Simulation, Université Saint-Quentin, Master 2 in HPC, 25–26);
- S. Errachi (Master Noyau Particule et Astroparticules, Univ. Paris Cité, 25–26);
- E. Larsson (Lund, Master 2, 24–25);
- Z. Khurij (Lund, Master 2, 24–25);
- **A. Storck** (Lund, Master 2, 23–24);
- A.-M. Söderman (Lund, Master, 23–24);
- **Z. Kocjan** (UCL, MSc, 21–23);
- J. Warbrick (UCL, MSci, 20–21);
- **E. Pharabod** (Polytechnique, France, Master 2, 20–21).

2016–19 **Teaching Assistant**

Sorbonne Université, Paris, France

Courses included: concept and methods of Physics at B.Sc. level (192 hours). Graded all written work, oral and final written exams and assisted with labs.

## Outreach activities

2019–now **Outreach presentations in high-schools, museums, for the general public, for open house days.**

2020–22 **Host and co-founder of the “Astronomy on Tap” London satellite**

Fortnightly general public online presentations ([online](#) due to the pandemic, more than 4,600 views). Awarded £1,000 by UCL Astronomy department to carry our activities.

2020 Scientific expertise to translate the general public book ‘A History of the Universe in 100 stars’.

2019 **Speaker at the “Pint of Science” festival**

Paris, France

2017–19 **Journée de la Science (Open House days)**

Sorbonne Université, France

Presented activities of the IAP, set up and performed hand-based experiments.

## Visiting programs, schools and conferences

Invited talks at conferences and seminars are listed below. Poster presentations are highlighted as “x”.

### — Invited talks

|         |                                                    |                                                              |
|---------|----------------------------------------------------|--------------------------------------------------------------|
| 06/2025 | National Astronomical Meeting                      | Durham University, Durham, United Kingdom                    |
| 06/2025 | Yt workshop                                        | Royal Astronomical Observatory, Edinburgh, UK                |
| 03/2023 | Connecting Galaxies to Cosmology visiting Program  | KITP, Santa Barbara, USA                                     |
| 10/2022 | 10th Workshop on Cosmology and Structure Formation | KIAS, Seoul, South Korea                                     |
| 03/2022 | Cosmic Cartography                                 | <i>online</i> , Kavli IPMU, Kashiwa, Japan                   |
| 01/2021 | ΛCDM: Dark Matter In Cosmology                     | <i>online</i> , Monthly meeting of London-based cosmologists |
| 11/2019 | Yonsei-IAP Workshop                                | <i>online</i>                                                |
| 03/2019 | Yt workshop                                        | University of Illinois, Urbana, USA                          |

### — Seminars

|         |                                                                                                                                                                 |                                                                                  |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 12/2025 | Seminar PKU/KIAA                                                                                                                                                | Peking University/Kavli Institute for Astronomy and Astrophysics, Beijing, China |
| 12/2025 | Colloquium Tsinghua University                                                                                                                                  | Tsinghua University, Beijing, China                                              |
| 06/2025 | Institute for Fundamental Physics of the Universe seminar                                                                                                       | IFPU, Trieste, Italy                                                             |
| 04/2025 | City University New York (CUNY) Physics Seminar                                                                                                                 | CUNY, New York, USA (online)                                                     |
| 04/2023 | Kavli Institute for Theoretical Physics blackboard talk                                                                                                         | KITP, Santa Barbara, USA                                                         |
|         | Prestigious talks intended to explain the science of one program to the other KITP program participants, locals, and scientists outside of a specialized field. |                                                                                  |
| 02/2022 | Berkeley Cosmology Seminar                                                                                                                                      | <i>online</i> , Berkeley, USA                                                    |
| 11/2021 | Oxford Cosmology Seminar                                                                                                                                        | Oxford, UK                                                                       |

### — Contributed talks

|         |                     |                                                |
|---------|---------------------|------------------------------------------------|
| 11/2025 | RAMSES SNO Meeting  | Université Paris Diderot, Paris, France        |
| 05/2025 | RAMSES User Meeting | Observatoire de Strasbourg, Strasbourg, France |
| 11/2024 | DYABLO Workshop     | IAP, France                                    |

|         |                                                                                 |                                                         |
|---------|---------------------------------------------------------------------------------|---------------------------------------------------------|
| 09/2024 | Framing the Big Picture of Galaxy Star Formation Quenching with JWST and Euclid | ESAC, Madrid, Spain                                     |
| 09/2024 | High- $z$ Kinematics Workshop III                                               | Max Planck Institute for Astronomy, Heidelberg, Germany |
| 03/2024 | Building Galaxies from Scratch                                                  | University of Vienna, Austria                           |
| 01/2024 | X D-LOCKS Meeting                                                               | Technical University of Denmark, Copenhagen, Denmark    |
| 12/2023 | New Simulations for New Problems in Galaxy Formation                            | Institut d'Astrophysique de Paris, France               |
| 08/2023 | Santa Cruz Galaxy Workshop                                                      | University of California Santa Cruz, USA                |
| 07/2022 | X National Astronomy Meeting (NAM)                                              | Warwick, UK                                             |
| 06/2022 | X EAS Meeting                                                                   | Valencia, Spain                                         |
| 06/2022 | Journées du PNCG (cosmology & galaxies)                                         | Observatoire Astronomique de Strasbourg, France         |
| 09/2021 | RAMSES User Meeting                                                             | online, Strasbourg Observatory, France                  |
| 07/2021 | Scipy 21: data analysis and code development in Python (900 participants)       | online                                                  |
| 12/2020 | RHyTHM: ResearchH using Yt Highlights Meeting.                                  | online                                                  |
| 11/2020 | KIAS Cosmology Workshop.                                                        | online                                                  |
| 10/2019 | KIAS Internal Workshop                                                          | KIAS, Seoul, South Korea                                |
| 09/2018 | West Coast Swings workshop                                                      | ICRAR, Perth, Australia                                 |
| 05/2018 | SPIN(E) ANR Meeting                                                             | ROE, Edinburgh, UK                                      |
| 09/2017 | SPIN(E) ANR Meeting                                                             | Agay, France                                            |
| 09/2017 | RAMSES User Meeting                                                             | Nice Observatory, Nice, France                          |
| 09/2016 | RAMSES User Meeting                                                             | CRAL, Lyon, France                                      |

#### — Journal clubs

|         |                                 |                                               |
|---------|---------------------------------|-----------------------------------------------|
| 12/2025 | Tsinghua Astronomy journal club | Tsinghua University, Beijing, China           |
| 12/2025 | Yonsei journal club             | Yonsei University, Seoul, South Korea         |
| 12/2021 | 'FLAT' talk                     | Durham, UK                                    |
| 11/2021 | Cosmology Journal Club          | IAP, Paris, France                            |
| 11/2021 | Astrophysics Journal Club       | Racah Institute of Physics, Jerusalem, Israel |
| 10/2021 | Galaxy Coffee                   | MPIA, Heidelberg, Germany                     |
| 09/2021 | Cambridge Cosmology Seminar     | online, Institute of Astronomy, Cambridge, UK |
| 12/2018 | Journal club & visiting program | Astrophysics Department, Oxford, UK           |
| 04/2018 | CRAL journal club               | CRAL, Lyon, France                            |
| 10/2017 | KIAS journal club               | KIAS, Seoul, South Korea                      |
| 04/2017 | CITA Journal Club               | CITA, Toronto, Canada                         |

## Publication list

I have submitted **21** articles as lead or co-lead author (**20** already published in MNRAS and A&A). I also contributed to **27** other articles. My papers have been cited **1112** times ( $h$ -index of 17 as of 27<sup>th</sup> May 2026), [source: NASA/ADS](#).

#### — Submitted articles

1. “MEGATRON: the impact of non-equilibrium effects and local radiation fields on the circumgalactic medium at cosmic noon”, **Cadiou**, Katz, Rey, Agertz, Blaizot, Cameron, Choustikov, Devriendt, Hauk, Jones, Kimm, Laseter, Martin-Alvarez, Matsumoto, Nyhagen, Pearce, Rodríguez Montero, Rosdahl, Rufo Pastor, Sanati, Saxena, Slyz, Stiskalek, Storck & Yee, *submitted*, [arXiv:2510.05667](#), (2025).
2. “MEGATRON: how the first stars create an iron metallicity plateau in the smallest dwarf galaxies”, Rey, Katz, **Cadiou**, Sanati, Agertz, Blaizot, Cameron, Choustikov, Devriendt, Hauk, Ji, Jones, Kimm, Laseter, Martin-Alvarez, Matsumoto, Pearce, Revaz, Rodriguez Montero, Rosdahl, Saxena, Slyz, Stiskalek, Storck, Veenema & Yee, *submitted*, [arXiv:2510.05232](#), (2025).
3. “MEGATRON: Reproducing the Diversity of High-Redshift Galaxy Spectra with Cosmological Radiation Hydrodynamics Simulations”, Katz, Rey, **Cadiou**, Agertz, Blaizot, Cameron, Choustikov, Devriendt, Hauk, Jones,

## — Published articles

1. “On the Origin of Intracluster Light Based on the High-resolution Simulation, NEWCLUSTER”, Jeon, Contini, Han, Rhee, Martin, Kim, Lee, Kimm, Pichon, Byun, Dubois, **Cadiou**, Jang & Yi, *The Astrophysical Journal*, *998*, *30*, (2026).
2. “MEGATRON: the environments of Population III stars at Cosmic Dawn and their connection to present-day galaxies”, Storck, Katz, Devriendt, Slyz, **Cadiou**, Choustikov, Rey, Saxena, Agertz & Kimm, *Monthly Notices of the Royal Astronomical Society*, *548*, *stag529*, (2026).
3. “MEGATRON: Disentangling Physical Processes and Observational Bias in the Multi-Phase ISM of High-Redshift Galaxies”, Choustikov, Katz, Cameron, Saxena, Devriendt, Slyz, Rey, **Cadiou**, Blaizot, Kimm, Laseter, Matsumoto & Rosdahl, *The Open Journal of Astrophysics*, *9*, *58199*, (2026).
4. “Introducing NewCluster: First half of the history of a high-resolution cluster simulation”, Han, Yi, Dubois, Rhee, Jeon, Jang, Byun, **Cadiou**, Kim, Kimm & Pichon, *Astronomy and Astrophysics*, *705*, *A169*, (2026).
5. “The Impact of Star Formation and Feedback Recipes on the Stellar Mass and Interstellar Medium of High-Redshift Galaxies”, Katz, Rey, **Cadiou**, Kimm & Agertz, *The Open Journal of Astrophysics*, *9*, *56097*, (2026).
6. “EDGE: the emergence of dwarf galaxy scaling relations from cosmological radiation-hydrodynamics simulations”, Rey, Taylor, Gray, Kim, Andersson, Pontzen, Agertz, Read, **Cadiou**, Yates, Orkney, Scholte, Saintonge, Breneman, McQuinn, Muni & Das, *Monthly Notices of the Royal Astronomical Society*, *541*, *1195*, (2025).
7. “RAMSES-yOMP: Performance Optimizations for the Astrophysical Hydrodynamic Simulation Code RAMSES”, Han, Dubois, Lee, Kim, **Cadiou** & Yi, *The Astrophysical Journal*, *978*, *96*, (2025).
8. “Exploring the causal effect of cosmic filaments on dark matter haloes”, Storck, **Cadiou**, Agertz & Galárraga-Espinosa, *Monthly Notices of the Royal Astronomical Society*, *539*, *487*, (2025).
9. “EDGE-INFERNO: Simulating Every Observable Star in Faint Dwarf Galaxies and Their Consequences for Resolved-star Photometric Surveys”, Andersson, Rey, Pontzen, **Cadiou**, Agertz, Read & Martin, *The Astrophysical Journal*, *978*, *129*, (2025).
10. “How complex are galaxies? A non-parametric estimation of the intrinsic dimensionality of wide-band photometric data”, **Cadiou**, Laigle & Agertz, *Monthly Notices of the Royal Astronomical Society*, *537*, *1869*, (2025).
11. “Running with the bulls: The frequency of star-disc encounters in the Taurus star-forming region”, Winter, Benisty, Shuai, D  chene, Cuello, Anania, **Cadiou** & Joncour, *Astronomy and Astrophysics*, *691*, *A43*, (2024).
12. “The AGORA High-resolution Galaxy Simulations Comparison Project. IV. Halo and Galaxy Mass Assembly in a Cosmological Zoom-in Simulation at  $z \leq 2$ ”, Roca-F  brega, Kim, Primack, Jung, Genina, Hausammann, Kim, Lupi, Nagamine, Powell, Revaz, Shimizu, Strawn, Vel  zquez, Abel, Ceverino, Dong, Quinn, Shin, Segovia-Otero, Agertz, Barrow, **Cadiou**, Dekel, Hummels, Oh, Teyssier & AGORA Collaboration, *The Astrophysical Journal*, *968*, *125*, (2024).
13. “Probing cosmology via the clustering of critical points”, Shim, Pichon, Pogosyan, Appleby, **Cadiou**, Kim, Kraljic & Park, *Monthly Notices of the Royal Astronomical Society*, *528*, *1604*, (2024).
14. “Hot gas accretion fuels star formation faster than cold accretion in high-redshift galaxies”, Kocjan, **Cadiou**, Agertz & Pontzen, *Monthly Notices of the Royal Astronomical Society*, *534*, *918*, (2024).
15. “Estimating major merger rates and spin parameters ab initio via the clustering of critical events”, **Cadiou**, Pichon-Pharabod, Pichon & Pogosyan, *Monthly Notices of the Royal Astronomical Society*, *531*, *1385*, (2024).
16. “Evolution of cosmic filaments in the MTNG simulation”, Gal  rraga-Espinosa, **Cadiou**, Gouin, White, Springel, Pakmor, Hadzhiyska, Bose, Ferlito, Hernquist, Kannan, Barrera, Maria Delgado & Hern  ndez-Aguayo, *Astronomy and Astrophysics*, *684*, *A63*, (2024).
17. “pynbody/genetIC: Version 1.5.0”, Pontzen, **Cadiou**, svstopyra, nroth0815, Rey & rc-softdev-admin, –999, (2024).
18. “Hot gas accretion fuels star formation faster than cold accretion in high redshift galaxies”, Kocjan, **Cadiou**, Agertz & Pontzen, *American Astronomical Society Meeting Abstracts #243*, *243*, *306.02*, (2024).
19. “pynbody/tangos: Version 1.9.1”, Pontzen, Tremmel, **Cadiou**, Rey, Wright, Davies, philosoph & Quinn, –999, (2023).
20. “pynbody/pynbody: Version 1.5.2”, Pontzen, Ro  kar, **Cadiou**, Stinson, Mastropietro, Rey, Keller, Duffy, mkrets, Tremmel, Davies, Franck, Quinn, Sarmiento, Bovy, nroth0815, Coles, Ji, Applebaum, Zana, Biernacki, Herpich, mihaimt, Woods, EthTay, Altay, Winkler, Shaw & Moon, –999, (2023).



21. “Yr-project/yt\_astro\_analysis: yt\_astro\_analysis-1.1.3”, Smith, Turk, ZuHone, Robert, Skory, Hummels, Myers, Kowalik, Eganhila, Skillman, Warren, [Cadiou](#), Gsiisg, Wise, Madcpf, Leitner, Scopatz, De Val-Borro, Stark, Meng-Yuan, Keller, Dong, Richardson, Krafczyk, Goldbaum, Sankar & Stonnes, –999, (2023).
22. “Stellar angular momentum can be controlled from cosmological initial conditions”, [Cadiou](#), Pontzen & Peiris, *Monthly Notices of the Royal Astronomical Society*, 517, 3459, (2022).
23. “Forecasts for WEAVE-QSO: 3D clustering and connectivity of critical points with Lyman- $\alpha$  tomography”, Kraljic, Laigle, Pichon, Peirani, Codis, Shim, [Cadiou](#), Pogosyan, Arnouts, Pieri, Iršič, Morrison, Oñorbe, Pérez-Ràfols & Dalton, *Monthly Notices of the Royal Astronomical Society*, 514, 1359, (2022).
24. “Gravitational torques dominate the dynamics of accreted gas at  $z > 2$ ”, [Cadiou](#), Dubois & Pichon, *Monthly Notices of the Royal Astronomical Society*, 514, 5429, (2022).
25. “Matplotlib label lines”, [Cadiou](#), –999, (2022).
26. “Matplotlib label lines”, [Cadiou](#), –999, (2022).
27. “FyeldGenerator”, [Cadiou](#), –999, (2022).
28. “On the causal origin of the angular momentum of dark matter halos and galaxies”, [Cadiou](#), EAS2022, European Astronomical Society Annual Meeting, 476, (2022).
29. “pynbody/genetIC: Version 1.3.5”, Pontzen, [Cadiou](#), Svstopyra, Nroth0815, Rey & Rc-Softdev-Admin, –999, (2022).
30. “The causal effect of environment on halo mass and concentration”, [Cadiou](#), Pontzen, Peiris & Lucie-Smith, *Monthly Notices of the Royal Astronomical Society*, 508, 1189, (2021).
31. “Angular momentum evolution can be predicted from cosmological initial conditions”, [Cadiou](#), Pontzen & Peiris, *Monthly Notices of the Royal Astronomical Society*, 502, 5480, (2021).
32. “The clustering of critical points in the evolving cosmic web”, Shim, Codis, Pichon, Pogosyan & [Cadiou](#), *Monthly Notices of the Royal Astronomical Society*, 502, 3885, (2021).
33. “EDGE: a new approach to suppressing numerical diffusion in adaptive mesh simulations of galaxy formation”, Pontzen, Rey, [Cadiou](#), Agertz, Teyssier, Read & Orkney, *Monthly Notices of the Royal Astronomical Society*, 501, 1755, (2021).
34. “Tracing the simulated high-redshift circumgalactic medium with Lyman  $\alpha$  emission”, Mitchell, Blaizot, [Cadiou](#), Dubois, Garel & Rosdahl, *Monthly Notices of the Royal Astronomical Society*, 501, 5757, (2021).
35. “The OBELISK simulation: Galaxies contribute more than AGN to H I reionization of protoclusters”, Trebitsch, Dubois, Volonteri, Pfister, [Cadiou](#), Katz, Rosdahl, Kimm, Pichon, Beckmann, Devriendt & Slyz, *Astronomy and Astrophysics*, 653, A154, (2021).
36. “pynbody/genetIC: Version 1.3”, Pontzen, [Cadiou](#), Svstopyra, Nroth0815, Rey & Rc-Softdev-Admin, –999, (2021).
37. “pynbody/genetIC: Version 1.2”, Pontzen, Svstopyra, [Cadiou](#), Nroth0815, Rey & Rc-Softdev-Admin, –999, (2021).
38. “The clustering of critical points in the evolving cosmic web”, Shim, Codis, Pichon, Pogosyan & [Cadiou](#), *Bulletin of Korean Astronomical Society*, 46, 47.2, (2021).
39. “When do cosmic peaks, filaments, or walls merge? A theory of critical events in a multiscale landscape”, [Cadiou](#), Pichon, Codis, Musso, Pogosyan, Dubois, Cardoso & Prunet, *Monthly Notices of the Royal Astronomical Society*, 496, 4787, (2020).
40. “pynbody/pynbody: Version 1.0.2”, Pontzen, Roškar, Stinson, [Cadiou](#), Keller, Duffy, Mkrets, Tremmel, Mastropietro, Sarmiento, Quinn, Nroth0815, Coles, Ji, Biernacki, GFG-CHAOS, Herpich, Mihaimt, Woods, Bovy, Emapple, Altay, De Val-Borro, Shaw, Moon, TobiBu, Mueslo & Perret, –999, (2020).
41. “Dense gas formation and destruction in a simulated Perseus-like galaxy cluster with spin-driven black hole feedback”, Beckmann, Dubois, Guillard, Salome, Olivares, Polles, [Cadiou](#), Combes, Hamer, Lehnert & Pineau des Forets, *Astronomy and Astrophysics*, 631, A60, (2019).
42. “Accurate tracer particles of baryon dynamics in the adaptive mesh refinement code RAMSES”, [Cadiou](#), Dubois & Pichon, *Astronomy and Astrophysics*, 621, A96, (2019).
43. “Galaxies flowing in the oriented saddle frame of the cosmic web”, Kraljic, Pichon, Dubois, Codis, [Cadiou](#), Devriendt, Musso, Welker, Arnouts, Hwang, Laigle, Peirani, Slyz, Treyer & Vibert, *Monthly Notices of the Royal Astronomical Society*, 483, 3227, (2019).
44. “Galaxy evolution in the metric of the cosmic web”, Kraljic, Arnouts, Pichon, Laigle, de la Torre, Vibert, [Cadiou](#), Dubois, Treyer, Schimd, Codis, de Lapparent, Devriendt, Hwang, Le Borgne, Malavasi, Milliard, Musso, Pogosyan, Alpaslan, Bland-Hawthorn & Wright, *Monthly Notices of the Royal Astronomical Society*, 474, 547, (2018).
45. “How does the cosmic web impact assembly bias?”, Musso, [Cadiou](#), Pichon, Codis, Kraljic & Dubois, *Monthly Notices of the Royal Astronomical Society*, 476, 4877, (2018).